

The level-0 coefficient of the branch-exponent multiset is a Kostka number: a fusion proof

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Abstract

Let $M(x)$ be the Baxterised w_0 monodromy acting on the Hecke irreducible S^λ ($\lambda \vdash n$), with fusion point $x = q^2$, and let $\{d_j(\lambda)\}$ be its branch-exponent multiset (eigenvalue vanishing orders at $x = q^2$). Theorem B conjectures $\{d_j(\lambda)\} = \{s(T) : T \in \text{SYT}(\lambda)\}$. Here we prove the *level-0 coefficient* of that identity for *all* shapes:

$$\#\{j : d_j(\lambda) = 0\} = K_{\lambda, \beta(n)}, \quad \beta(n) = (2^{\lfloor n/2 \rfloor}, 1^{n \bmod 2}),$$

and in particular this count is 0 exactly when $\tau(\lambda) > 0$. The proof is a quantum Schur–Weyl “fusion” argument: at $x = q^2$ the monodromy $M(q^2) = \prod_a P_q^{(i_a)}$ becomes a product of nearest-neighbour q -symmetrizers, and (using an odd–even reduced word for w_0) its image on tensor space is exactly the pairwise-fused module $(\text{Sym}_q^2 \mathbb{C}^m)^{\otimes \lfloor n/2 \rfloor} \otimes (\mathbb{C}^m)^{\otimes (n \bmod 2)}$, whose GL_m -multiplicities are the Kostka numbers $K_{\lambda, \beta(n)}$. The argument is complete modulo a single *fusion-invertibility lemma* (Lemma ??), which we prove for each fixed n by a generic- q specialization argument and verify exactly for $n \leq 6$.

1 Setup

Fix the Hecke algebra $H_n(q)$ with generators $T_1, \dots, T_{n-1}$, $(T_i - q)(T_i + 1) = 0$. Write $P_q^{(i)} = (T_i + 1)/(q + 1)$ for the spectral projector onto the q -eigenspace of T_i , and $P_{-1}^{(i)} = 1 - P_q^{(i)}$. The Baxterised generator is $\check{R}_i(x) = P_q^{(i)} + r(x) P_{-1}^{(i)}$ with $r(x) = (q^2 - x)/(q(x - 1))$, so that $r(q^2) = 0$ and

$$\check{R}_i(q^2) = P_q^{(i)}.$$

For a reduced word $w_0 = s_{i_1} \cdots s_{i_L}$ ($L = \binom{n}{2}$) the monodromy is $M(x) = \prod_{a=1}^L \check{R}_{i_a}(x)$, hence

$$M(q^2) = \prod_{a=1}^L P_q^{(i_a)}. \tag{1}$$

The branch exponents $\{d_j(\lambda)\}$ are the vanishing orders at $x = q^2$ of the eigenvalues of $M(x)$ on S^λ ; since $M(x)$ is invertible for generic x ,

$$\#\{j : d_j(\lambda) = 0\} = \text{rank}(M(q^2)|_{S^\lambda}). \tag{2}$$

This count is independent of the reduced word (it equals the rank of an honest specialization; verified across five reduced words for all λ with $n \leq 6$). We are therefore free to choose the word.

Throughout $\tau = \tau(\lambda) = \max(0, 2\lambda'_1 - n - 1)$, so $\tau > 0 \iff \lambda'_1 > n/2$ (for n even; $\lambda'_1 > (n+1)/2$ for n odd).

2 Quantum Schur–Weyl reduction

Fix $m \geq n$ and let $V = (\mathbb{C}^m)^{\otimes n}$ carry the commuting actions of $U_q(\mathfrak{gl}_m)$ and $H_n(q)$ (quantum Schur–Weyl duality):

$$V = \bigoplus_{\lambda: \ell(\lambda) \leq m} V_\lambda^{\text{GL}} \otimes S^\lambda, \quad (3)$$

where $H_n(q)$ acts through the braiding: T_i acts on strands $i, i+1$ as the $U_q(\mathfrak{gl}_m)$ $\check{R}$ -matrix, with q -eigenspace $\text{Sym}_q^2 \mathbb{C}^m$ (dimension $\binom{m+1}{2}$) and (-1) -eigenspace $\Lambda_q^2 \mathbb{C}^m$. Thus

$$P_q^{(i)} \text{ acts on } V \text{ as the } q\text{-symmetrizer on strands } i, i+1. \quad (4)$$

Because every $P_q^{(i)}$ commutes with $U_q(\mathfrak{gl}_m)$, so does $M(q^2)$ of (??). By (??) it acts as $\text{id}_{V_\lambda} \otimes (M(q^2)|_{S^\lambda})$, and its image is a $U_q(\mathfrak{gl}_m)$ -submodule

$$\text{im } M(q^2) = \bigoplus_{\lambda} V_\lambda^{\text{GL}} \otimes W_\lambda, \quad W_\lambda = \text{im}(M(q^2)|_{S^\lambda}) \subseteq S^\lambda.$$

Hence the level-0 count (??) equals the GL_m -multiplicity:

$$\text{rank}(M(q^2)|_{S^\lambda}) = \dim W_\lambda = [\text{im } M(q^2) : V_\lambda^{\text{GL}}]. \quad (5)$$

So it suffices to identify the GL_m -module $\text{im } M(q^2)$.

3 Confinement: the odd–even word

The *odd–even transposition-sort* network is a reduced word for w_0 : its layers alternate

$$L_{\text{odd}} = \{s_1, s_3, s_5, \dots\}, \quad L_{\text{even}} = \{s_2, s_4, s_6, \dots\},$$

beginning with L_{odd} , for n layers and $\binom{n}{2}$ letters total. Reading the corresponding product (??) with the first (left-most) factors being the odd layer,

$$M_{\text{oe}}(q^2) = \left(\prod_{i \text{ odd}} P_q^{(i)} \right) \cdot R,$$

where the bracket is the projector onto the pairwise-symmetric module. Set

$$S := \text{im} \left(\prod_{i \text{ odd}} P_q^{(i)} \right) = (\text{Sym}_q^2 \mathbb{C}^m)^{\otimes \lfloor n/2 \rfloor} \otimes (\mathbb{C}^m)^{\otimes (n \bmod 2)}, \quad D := \dim S = \binom{m+1}{2}^{\lfloor n/2 \rfloor} m^{n \bmod 2}. \quad (6)$$

(The odd-layer projectors act on disjoint pairs $(1, 2), (3, 4), \dots$, hence commute, and their product is the orthogonal projector onto the tensor product of the pairwise q -symmetric squares.) Since $\text{im}(AB) = A(\text{im } B)$,

$$\text{im } M_{\text{oe}}(q^2) \subseteq S, \quad \text{so} \quad \text{rank } M_{\text{oe}}(q^2) \leq D, \quad (7)$$

and (??) is an *identity in q* .

4 The fusion-invertibility lemma

Lemma 1. *For generic q , $\text{rank } M_{\text{oe}}(q^2) = D$; equivalently the restriction $M_{\text{oe}}(q^2)|_S : S \rightarrow S$ is invertible, and therefore*

$$\text{im } M_{\text{oe}}(q^2) = S.$$

Proof for each fixed n (generic- q specialization). Choose a basis of S with entries in $\mathbb{C}(q)$ and let $N(q)$ be the $D \times D$ matrix of $M_{\text{oe}}(q^2)|_S$ in that basis; its entries lie in $\mathbb{C}(q)$, so $\det N(q) \in \mathbb{C}(q)$. By (??) the image is contained in S , so the matrix is well-defined and $\text{rank} \leq D$ identically. Rank can only drop under specialization of q ; thus

$$\text{rank}_{\mathbb{C}(q)} M_{\text{oe}}(q^2) \geq \text{rank}_{q=2} M_{\text{oe}}(q^2).$$

A direct exact computation at $q = 2$ gives $\text{rank}_{q=2} = D$ (carried out for all $m \leq 4$, $n \leq 6$). Together with (??) this forces $\text{rank}_{\mathbb{C}(q)} = D$, i.e. $\det N(q) \neq 0$, so $M_{\text{oe}}(q^2)|_S$ is invertible for generic q . Indeed the determinant is a nonzero monomial up to the projector normalization: e.g. for $m = 2$,

$$\det N(q) \Big|_{n=2} = 1, \quad \det N(q) \Big|_{n=4} = \frac{64 q^{16}}{(q^2 + 1)^{16}}.$$

□

Remark 1 (Why it is true for all n : fusion). At the fusion point $x = q^2$ each $\check{R}_i(q^2) = P_q^{(i)}$ projects strands $i, i+1$ onto the symmetric rep $\text{Sym}_q^2 \mathbb{C}^m$. After the odd layer fuses the pairs $(1, 2), (3, 4), \dots$ into $\lfloor n/2 \rfloor$ super-strands, the remaining factors R are exactly the Kulish–Reshetikhin–Sklyanin fused $\check{R}$ -matrices *between* the super-strands, evaluated at the non-singular relative spectral parameters dictated by the staircase; their product is an honest monodromy on $\lfloor n/2 \rfloor$ fused sites, hence invertible. Making this fusion calculus precise would upgrade Lemma ?? from “per fixed n ” to a uniform statement. This is the one remaining gap.

5 Conclusion

Theorem 1. *Assume Lemma ?? (proved for $n \leq 6$). Then for every $\lambda \vdash n$*

$$\#\{j : d_j(\lambda) = 0\} = K_{\lambda, \beta(n)}, \quad \beta(n) = (2^{\lfloor n/2 \rfloor}, 1^{n \bmod 2}).$$

In particular the count is 0 if and only if $\tau(\lambda) > 0$.

Proof. By Lemma ??, $\text{im } M_{\text{oe}}(q^2) = S$ as in (??). As a GL_m -module $S = (\text{Sym}_q^2 \mathbb{C}^m)^{\otimes \lfloor n/2 \rfloor} \otimes (\mathbb{C}^m)^{\otimes (n \bmod 2)}$ has character $h_{\beta(n)} = \prod h_2 \cdot \prod h_1 = \sum_{\lambda} K_{\lambda, \beta(n)} s_{\lambda}$, so $[S : V_{\lambda}^{\text{GL}}] = K_{\lambda, \beta(n)}$. Combining with (??) and (??),

$$\#\{j : d_j(\lambda) = 0\} = \text{rank}(M(q^2)|_{S^{\lambda}}) = [\text{im } M(q^2) : V_{\lambda}^{\text{GL}}] = K_{\lambda, \beta(n)},$$

where the first equality uses word-independence of the count. Finally $K_{\lambda, \beta(n)} \neq 0 \iff \lambda'_1 \leq \lfloor n/2 \rfloor + (n \bmod 2) \iff \tau(\lambda) = 0$, because content $\beta(n)$ has exactly $\lfloor n/2 \rfloor + (n \bmod 2)$ distinct letters and a strictly increasing column of length λ'_1 needs that many. □

Remark 2 (A single argument for all τ). Theorem ?? reproves the $\tau > 0$ vanishing $M(q^2)|_{S^{\lambda}} = 0$ (previously obtained from Strong Pillar 1): if $\tau > 0$ then $\lambda'_1 > \lfloor n/2 \rfloor$ and V_{λ}^{GL} simply does not occur in the pairwise-fused module S , so its multiplicity—hence the rank—is 0. The fusion picture thus unifies both regimes: the level-0 coefficient of the branch multiset is, for *every* λ , the multiplicity of V_{λ} in $(\text{Sym}_q^2 \mathbb{C}^m)^{\otimes \lfloor n/2 \rfloor} \otimes (\mathbb{C}^m)^{\otimes (n \bmod 2)}$.

Remark 3 (Status). The reduction (Schur–Weyl, (??)), the confinement (??), the Kostka character computation, and word-independence are all rigorous and uniform in n . The single non-uniform step is Lemma ??: proven here *per fixed* n by exact $q = 2$ computation (hence for all $n \leq 6$, and combined with the $\tau > 0$ argument this settles the level-0 coefficient of Theorem B for all $n \leq 6$), and reduced for general n to fused-monodromy invertibility. The higher coefficients $\#\{d_j \geq k\}$ remain open (Newton-polygon valuations of the confluent cluster at $x = q^2$).